Digital Twins for Cyber-Resilient Power Grids: A Structured Corpus Review of Defensive and Attack-Side AI, with a Research Agenda

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Abstract

Digital twins are becoming a reference architecture for monitoring, protecting and controlling modern power grids, and machine learning is being adopted inside those twins quickly. We review whether the security literature that names a grid digital twin and a security term treats AI as a defensive tool more often than it engages AI-enabled threats, or the twin itself, as an attack target. From twelve OpenAlex core queries we screened 1,287 unique records and retained a *ranked* top-100 subset (2020–2026) under a non-probabilistic rule favouring relevance, citations and recency; 83 studies were coded at full text across 46 fields and 17 at abstract level. Four further adversarial-AI probe queries widened the candidate pool to roughly 2,000 records; the released record does not establish whether probe-only records entered the retained corpus, so prevalence figures describe this corpus, not the wider literature. Machine learning is used in 70 of 83 full-text studies—for detection in 55; forecasting (12) and control (3) are not security functions. 35 of 83 (42%) engage any AI-enabled threat and 11 (13%) engage one substantively; within the 29 security-primary, non-review, detection-ML studies, 16 (55%) engage any and 6 (21%) substantively. No grid study treats the twin itself as the adversary's objective (the single strict *target* record is a vehicle-twin review outside the grid domain); 56 of 83 name an attack on the twin's data path or its learned components, treating the twin as a protected asset rather than a studied target. Threat coverage, as coded, concentrates at operation (66/83, on the field with the lowest interpretable agreement, $\kappa = 0.32$). In a 20-record double-coded sample (24% of the full-text records; both coders are authors of this review), agreement was $\kappa = 0.60$ (bootstrap 95% CI 0.20–0.90) for any engagement and $\kappa = 0.14$ (-0.23 to 0.64) for substantive

engagement; a misclassification sensitivity anchored on the second coder's coding leaves the headline near-unchanged (36 of 83, 95% CI 21–51). We close with a five-item research agenda in which every item is tied to the gap it addresses, the evidence that would confirm it, and the evidence that would disconfirm it.

Keywords: digital twin, power grid, cyber-physical security, adversarial machine learning, structured review, research agenda

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1. Introduction

Electric power grids are digitising faster than they are being secured. Wide-area measurement, distributed energy resources, electric-vehicle charging and market-coupled control have turned the grid into a cyber-physical system whose behaviour is increasingly mediated by software [1, 2]. The digital twin—a synchronised, bidirectional virtual counterpart of a physical asset, updated from live sensor streams—has moved in the same decade from concept to a reference architecture for grid monitoring, protection and control [3, 4]. That same synchronisation is what makes the twin valuable and what makes it a target.

Artificial intelligence (AI) has followed the twin into the grid. Machine learning now performs state estimation, anomaly detection, load and attack forecasting, and increasingly control [5, 6]. The security community has been quick to adopt it. The question this review asks is whether the same community has been as quick to model AI—and the twin itself—as something an adversary can attack. Throughout, 'digital twin' must be named by the authors, so adversarial-ML work on grid security that does not mention a twin is out of scope and is not represented in any figure here.

Early reading suggested a specific, testable asymmetry: grid digital-twin security research uses AI as a defensive tool far more often than it engages AI, or the twin, as an attack target. The hypothesis was formed from early reading of part of the same corpus, before coding began and with no prespecified confirmation threshold; the analysis is therefore exploratory, and the claim it supports is one of *emphasis*, not of *rate*: we make no temporal "adopts faster" claim, because the corpus is a recency-weighted ranked subset in which a year-wise comparison is confounded by selection (Section 4). This review quantifies that asymmetry across a screened corpus and turns it into a research agenda. It is a theory-testing structured review, not an empirical

deployment study.

1.1. Positioning against prior reviews

The closest work, and the reason this review had to be repositioned, is Habeeb, Weli and Almaliki's 2026 PRISMA review of "Dual-Role Digital Twins" in cyber-physical security [7]. Their contribution is real and prior: they name the dual role—the twin as both an expanded attack surface and a defence enabler—and set out four research directions around it. Their corpus is 20 studies, gathered with a query that has no specific attack operator, and spread across manufacturing, healthcare, IIoT, smart city and water management, with the power grid represented in roughly one study.

We therefore do not claim to have discovered the dual-role tension. Our position is narrower and complementary. Habeeb et al. name two axes; we measure a narrow, grid-side instance of the attack-surface axis—the twin as the adversary's explicit objective—with a published corpus, a closed threat vocabulary, and reported coder reliability. We additionally report looser tiers (the twin as an attacked data path or model, and the twin as a protected asset), because the strict-objective tier alone cannot speak to the broader attack-surface axis. In other words, they build the theory; we report a lower bound on its narrowest grid manifestation, not a test of the full axis. Section 3 sets out the mapping in full, including the five attack vectors their review identifies and our corpus's coverage of them.

1.2. Contributions

1. A screened corpus of 100 studies on digital twins for power-grid cybersecurity, drawn from 1,287 unique records of twelve OpenAlex core queries by a *declared, non-probabilistic* top-100 ranking (relevance, citations, recency). The corpus is a ranked subset of 492 eligible records, not a census or a random sample; the ranking and its biases are reported with the corpus (Section 4).
2. A 46-field extraction of that corpus, coded by computer-assisted human reading, with the extraction method disclosed rather than hidden (Section 4).
3. A two-coder reliability study on a 20-record blind sample (24% of the 83 full-text records), reported field by field with κ, bootstrap confidence intervals, PABAK and bias indices, including the fields where agreement is only fair (Section 4.8).

4. The headline measurement, with matched denominators: machine learning is used in 70/83 full-text studies (detection in 55); 35/83 engage any AI-enabled threat and 11/83 engage one substantively; within the 29 security-primary, non-review, detection-ML studies the figures are 16/29 and 6/29. No grid study treats the twin as the adversary's objective; 56/83 name an attack channel on the twin's data path or models (Section 5.5).
5. A mapping from our threat vocabulary to the five vectors of Habeeb et al. [7], presented as vocabulary-level coverage only—a single-author judgment, not reliability-checked evidence that the vectors are dealt with empirically (Section 3).
6. A five-item research agenda in which every item is tied to the gap that justifies it, the evidence that would validate it, and the evidence that would falsify it (Section 7).

1.3. Research questions

RQ1 How is the digital twin positioned in grid cybersecurity research—as a defensive tool, an attack target, or both?

RQ2 Which threats are named, at which grid domains and lifecycle stages, and with what threat models?

RQ3 How far does the corpus engage AI-enabled threats (adversarial machine learning, evasion, poisoning, LLM-specific, agentic)?

RQ4 How rigorous is the evidence base—evaluation method, maturity, baselines, metrics, and released code or data?

RQ5 What are the resulting gaps, and what would close them?

1.4. Structure

Section 2 gives the technical background. Section 3 reviews prior work and positions this review against it. Section 4 is the protocol. Section 5 reports results, figure-led, across RQ1–RQ4. Section 6 discusses them. Section 7 sets the research agenda for RQ5. Section 8 is the threats-to-validity analysis, and Section 9 concludes.

2. Background

2.1. Digital twins in power systems

A digital twin is a virtual representation of a physical entity that is connected to it by a bidirectional data link, such that the virtual state tracks the physical state and, in the stronger formulations, the physical entity can be actuated through the twin. Three properties distinguish a twin from an ordinary model or simulation: synchronisation with live data, bidirectionality, and asset specificity. Grid applications span state estimation and observability, protection and control, fault and attack diagnosis, asset-health monitoring, and training and planning [3–5].

The maturity of grid twins varies widely. Some reported twins are conceptual architectures with no implementation; many are prototypes demonstrated on simulation or testbed data; a small number report validation against real grid assets [6, 8]. This maturity gradient matters for security, because a threat model that is plausible on a conceptual twin may not be reachable on a real deployment, and vice versa.

2.2. Cyber threats to grids and to twins

The classical grid threat model is adversarial and studied: false data injection against state estimation, replay, denial of service, man-in-the-middle on telemetry links, false command injection, and supply-chain compromise [1, 2]. These are conventional cyber-physical attacks and are well represented in the corpus (Section 5.4).

The twinning layer adds attack surface that the classical model does not cover. The bidirectional link can be exploited in both directions; the twin synchronises with sensor streams, so any data-poisoning channel into those streams reaches the twin; and where the twin uses learned components—detectors, forecasters, controllers—those components can themselves be attacked by distribution-shift, evasion, or training-time poisoning. This is the attack-surface axis that Habeeb et al. name [7].

2.3. AI on both sides of the boundary

AI enters grid security from two directions. Defensively, machine learning performs intrusion and anomaly detection, attack classification, load and attack forecasting, and increasingly control. Offensively, the same model families are themselves vulnerable: adversarial examples can evade learned

detectors, poisoning can corrupt training data drawn from compromised sensor streams, and large language models introduce prompt-level and social-engineering vectors that did not exist in earlier grids. An "AI-enabled threat" in this review means an attack in which the adversary uses AI, or attacks an AI component, or both—as distinct from a conventional cyber-physical attack that happens to be detected by an AI method. These two directions have different implications (an AI-wielded attack changes the adversary; an attack on an AI component changes the victim), and the headline construct *combines* them. The coding does not separate the two directions per record; Section 5.5 reports a notes-based separation for the substantive studies and flags the lumping as a construct limitation (Section 8).

A brief terminology note. 'Digital twin' has largely superseded 'CPS model' and 'HIL simulation' in recent grids literature, so papers that functionally twin a grid without using the phrase are out of scope of the search, despite carrying relevant threat models.

3. Related work and positioning

3.1. The closest review

Habeeb, Weli and Almaliki (2026, *Digital Twin*) review Dual-Role Digital Twins for cyber-physical security with PRISMA 2020: 412 identified, 318 screened, 72 eligibility-checked, 20 included, spanning January 2019 to December 2025 [7]. They ground the dual role in Complex Adaptive Systems, functional duality, and zero-trust architecture, and they name five attack vectors. Their search string is `("Digital Twin" OR "DT") AND ("Cybersecurity" OR "Attack Surface" OR "Defense" OR "Honeypot" OR "Intrusion Detection")`. It contains no term for a specific attack (no adversarial machine learning, poisoning, evasion, or reverse-pathway operator), so attack-side records have to enter through the security and defence terms. The same limitation applies to our twelve core queries, which is why we added the four adversarial-AI probes.

3.2. The bridge: their vectors, our vocabulary

Our codebook's threat vocabulary was designed independently of their review; the claim rests on the dated codebook and design log in the repository and cannot be verified from this text alone. Table 1 maps each of their five attack vectors onto our vocabulary terms. The mapping shows *vocabulary-level coverage only*: every one of their vectors is expressible in our

Table 1: Our threat vocabulary mapped onto the five attack vectors of Habeeb et al. [7]. Counts are records naming the term (full text, $n = 83$). The mapping is vocabulary-level coverage only; it is a single-author judgment, was not double-coded, and is not evidence that any vector is dealt with empirically. `dos` and `fdi` are classical attacks placed under the vector whose surface they exploit. Constructs beyond their five vectors: `ai_llm` (4 records), `ai_agentic` (2).

#	Their vector	Our vocabulary terms	Records naming it
i	Bidirectional data-link exploitation	`mitm`, `replay`, `dos`[c]	26, 23, 49
ii	Adversarial data poisoning on sensor streams	`poisoning`, `fdi`[c]	22, 46
iii	AI/ML model exploitation via training-distribution attacks	`ai_adv_ml`, `ai_poisoning`, `ai_evasion`	24, 19, 9
iv	Reverse attack pathway (twin → plant)	`reverse_pathway`, `false_command`[c]	10, 22
v	Insider and supply-chain compromise of the twin	`insider`, `supply_chain`	20, 19

[c]Classical (non-AI) attack, mapped by exploited surface.

closed vocabulary. It is not evidence that the corpus deals with the vectors empirically, and it is a single-author judgment that was not double-coded.

The relationship between the two search designs is the mirror image of the mapping. Our search adds retrieval operators their query does not contain—adversarial machine learning, poisoning, evasion and large-language-model terms—so records whose visibility depends on those terms could enter our pool but not theirs. Two of our coded constructs, LLM-specific threats (4 records) and agentic attack automation (2), fall outside their five vectors altogether. The remaining added terms map onto their own vectors, which is why the table reports them as coverage rather than as novelty. The mapping also places two classical attacks, `dos` and `fdi`, under their nearest vectors (i and ii); those attacks are not AI-specific, and the placement follows the vector whose surface they exploit, not an identity of mechanism.

3.3. Other reviews in the corpus

Several corpus entries are themselves reviews [1–5, 9]. Their coverage is broad on the defensive side. Our contribution relative to them is the

quantified attack-side measurement: we do not merely note that AI threats matter, we count how often the corpus engages them and find that the answer is rarely, and rarely substantively.

4. Methodology

4.1. Design

This is a *structured* literature review reported in PRISMA-2020 flow format for transparency of search and selection. It is not a systematic review in the PRISMA sense: it searches one database (OpenAlex), has no registered protocol, and retains a ranked, non-random subset of the eligible records. We use PRISMA flow reporting because the alternatives (a narrative review) would hide the selection rule rather than expose it. The protocol was fixed before extraction, with the codebook revisions noted below. The review is theory-testing in intent: the asymmetry stated in the introduction is the hypothesis, and the retained 100 studies are the corpus on which it is explored—with no pre-specified confirmation threshold, as Section 1 states.

4.2. Search strategy

We searched OpenAlex with twelve core queries combining a digital-twin term, a grid-domain term, and a security term, yielding 1,287 unique records after de-duplication. We then ran four additional adversarial-AI probes—combining digital-twin and grid terms with adversarial machine learning, poisoning, evasion, and large-language-model terms—which raised the candidate pool to approximately 2,000. The probes exist because the core grid-security vocabulary is not expected to retrieve the attack-side literature.

What the released record does *not* establish is the probes' effect on the corpus. The screened set of 1,287 records derives from the twelve core queries; per-record query provenance for the probe-added records (roughly 700) was not retained, so we cannot report how many of the 492 eligible, or of the retained 100, are retrievable only through the probes. The retrieval-bias defence that the probes were designed to provide therefore cannot be made from these data, and we do not make it (Section 8, Search bias). Releasing the query strings, the run dates and the per-record provenance is a precondition for resolving this, and the repository now carries a provenance manifest (‘SEARCH_PROVENANCE.md‘) stating exactly what must be released; the query strings themselves were not preserved in the released record at the time of writing and are listed there as a required artifact.

4.3. Selection and inclusion

We screened the 1,287 core-query records on title and abstract. A record was eligible if its title or abstract contained at least one digital-twin term, at least one grid, power or energy term, and at least one cyber or security term; 795 records failed at least one of the three conditions, leaving 492 eligible.

The grid term is deliberately broad—*grid, power or energy*—and the eligibility rule therefore admits adjacent domains. Four retained records are outside the power-grid domain on a strict reading (a vehicle-twin threat-modelling review, a cyber-fraud study, a network-traffic anomaly-detection paper with no grid or twin instantiation, and a nuclear physical-protection risk analysis in which no twin is instantiated); three more are borderline (nuclear digital twins, battery twins for telecom supply, and a security-education cyber twin). A strict-scope sensitivity analysis that removes these records is reported alongside every headline figure (Section 5.5); the borderline records are listed there with their identifiers.

The eligible set was then narrowed to 100 by a non-probabilistic rule combining OpenAlex relevance score, citation count, and recency, with the final cut at rank 100. This ranking is a *selection* step, not an eligibility criterion, and it has known direction biases: it favours reviews (36% of the retained corpus) and established, cited defensive work, and penalises new, uncited attack-side work. Because the rule is not a random sample, every prevalence figure in Section 5 is a statement about the retained 100 studies and should not be read as an estimate of the wider literature; the corpus's 76% share of 2025–26 records is partly an artefact of the recency weight, not a finding about the field. Small denominators bound all percentages, and the uncertainty that a confidence interval should carry here is coding misclassification, which we quantify in Section 5.7 rather than with sampling intervals.

4.4. Screening

Screening was title-and-abstract, applying the three-term eligibility rule above: 795 of the 1,287 core-query records were excluded for missing at least one of the three terms (digital twin; grid, power or energy; cyber or security), leaving 492 eligible records. Inclusion then ranked the eligible set by relevance, citation count, and recency, and retained the top 100, as declared in the selection step above. Figure 1 is the flow. The flow reports the core-query pipeline; the approximately 700 probe-added records are shown

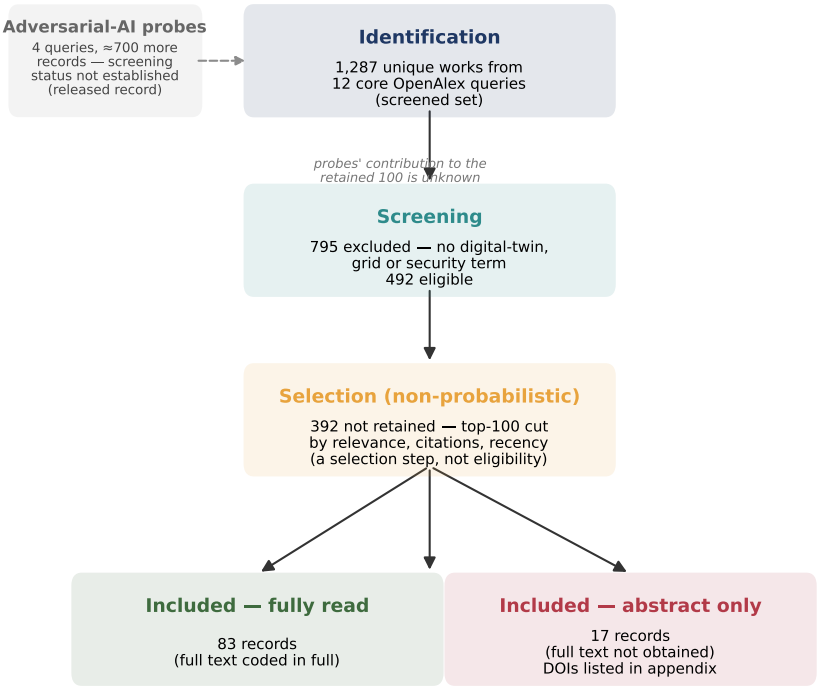


Figure 1: PRISMA-style flow. The 1,287 screened records derive from the twelve core queries; the four adversarial-AI probes added roughly 700 candidate records whose screening status is not established in the released record (Section 4). The top-100 cut is a declared non-probabilistic selection step, not an eligibility criterion. Eighty-three records are coded at full text and 17 at abstract level only, because their full text was not obtained.

at identification but their screening status is not established in the released record, as Section 4 discloses.

4.5. Extraction

We extracted 46 fields per record. Extraction was computer-assisted but human-coded: a script surfaced keyword hits and context snippets for each field, and every code was assigned by a human reading those snippets in context. Keyword snippets were scored against the full text and the surrounding paragraph, not the excerpt alone; reference lists and author biographies were stripped before scanning, so absence claims rest on the surviving body text.

The extraction is reproducible from the coded excerpts and DOIs, which are released.

Of the 100 records, 83 were available in full text and coded in full; 17 were available only as abstracts and were coded at abstract level only. All abstract-only fields that require full text are treated as missing, not as zero, throughout. The 17% non-retrieval rate is an access limitation of the authors (paywalled or blocked full texts), not a property of the literature or a reproducibility finding; the affected records are listed with DOIs so readers can see exactly what is missing (Appendix 9).

4.6. Codebook

The codebook defines each field and its controlled vocabulary, and was revised five times during extraction. The revisions are logged in the repository and are load-bearing:

- **R2** distinguishes a review from an enabling paper, and classifies attack-generating work as *bridge*.
- **R3** requires a named interface or protocol before `attack_surface_named` may be coded *yes*; passing mentions do not qualify.
- **R4** prevents passing mentions from triggering the *multi* lifecycle value.
- **R5** contracts `ml_family` to a controlled vocabulary.
- **R6** fixes evaluation-method precedence (*hardware-in-the-loop* > testbed > simulation) and maps surveys to *concept* maturity.

The `twin_role` field requires positive evidence. *both* needs explicit coverage of the defensive and attack-target roles *and* of the tension between them; *target* requires that the twin be the adversary's objective rather than an asset to be protected. *both* = 0 is therefore definitional: a paper cannot score *both* without stating the tension, and we report that zero as a statement about *explicit articulation*, not as evidence that dual-role practice is absent. The strict *target* rule is likewise one end of a tiered measurement: in a security setting an asset that needs protecting is an asset an adversary may attack, so Section 5.3 reports three tiers side by side—twin as the adversary's objective (strict), twin's data path or learned components named as an attack channel (interface and model threats from the closed vocabulary), and twin

as protected asset (*tool*). Candidates such as [10], "Enhancing the cybersecurity of battery digital twins," were read as *tool* because the twin remains an asset under protection; under the looser tiers that study counts as naming an attack channel, which is the reading its content supports.

4.7. Scoring of AI-threat engagement

Each AI-enabled threat field was coded on a 0–2 ordinal scale: 0 = not addressed; 1 = mentioned, listed, or peripheral (fewer than two sentences or no failure-mode detail); 2 = substantive (a threat model, mechanism, experiment, or mitigation specific to that vector, or a section dedicated to it). The threshold was applied to the full text rather than the abstract. `ai_threat_overall` is 1 if any vector is 1 or 2, and 2 if any vector is 2. Both coders coded the scale independently; per-vector agreement is reported in Section 5.7.

4.8. Reliability

A second coder independently coded 20 of the 83 full-text records (24% of the full-text corpus, drawn from the full-text frame only), blind to coder 1's values and to the screening label. Both coders are authors of this review; coder 1 is the codebook author and coded 80% of the corpus. The sample of 20 contains about 0.2 expected *target* cases at corpus prevalence, so `twin_role` agreement carries almost no information about the rare class that the headline needs; the same holds for the rarer AI vectors. Agreement is reported field by field in Section 5.7 as Cohen's κ with bootstrap 95% intervals, PABAK (prevalence-adjusted bias-adjusted kappa, which removes the prevalence depression that affects κ), and a bias index (the largest between-coder difference in category marginals), together with percentage agreement where a field is near-constant and κ is undefined. With $n = 20$, the large-sample standard error of κ is of the order of 0.2–0.3 even for the better fields; cells in Table 8 are point estimates, and a stable ranking of field difficulty is not achievable from this sample.

4.9. Adjudication

All field-level disagreements were adjudicated against the codebook and the underlying evidence. The rules, stated here rather than buried, are: adopt the value the codebook and evidence dictate; where both coders' values are defensible, coder 1's prevails; where neither fits, the codebook rule decides. Adjudication changed 20 net cells (Section 5.7). No headline

field moved, but under a tie-break that defaults to coder 1 this is the expected outcome, not evidence of robustness: headline fields could only have moved had the adjudicator overridden coder 1, and the adjudicator works with coder 1. Adjudication therefore does not test the headline; the released pre-adjudication codes are the honest artefact, and they are published frozen alongside the adjudicated matrix.

4.10. False-negative audit

Because the headline is an absence—of attack-side engagement—we audited it, and we report the audit with its design limits stated rather than implied. We ran a strict AI-threat lexicon over the 48 records coded as *not engaging* any AI threat and a broader eleven-family net. Both the extraction and the audit are keyword-assisted, so the audit shares the extraction's blind spot: "no engagement" partly means "no keyword hit", and asking a lexicon to re-find what a lexicon missed is close to circular. Its positive control recovers only 7 of the 11 substantive positives, and its recall on mentioned-only (score-1) engagement—the harder case, covering 24 of the 35 engaged records—was never measured. Of the 48 negatives, 38 fired the broad net and 23 fired a strong family; only the 23 strong firings were individually inspected, and the 15 weak firings are reported here as *not* individually inspected. The audit is retained as a disclosure of the false-negative channel, not as a bound on it; the design that would bound it—an independent full-text re-read of a random sample of the 48 negatives— is stated as required future work in Section 5.8.

4.11. Verification and reproducibility

Every DOI was verified against an authoritative registry: 96 of 97 resolved with a title match (95 exact, one near), one DOI is not registered, and three corpus records carry no DOI. Corpus metadata was cross-checked against Crossref, and 18 of 54 venue labels disagreed, which we resolved in favour of the registry. All extraction data, the codebook, the coding sheets, the reliability script, and the figure script are released as a reproducibility package.

5. Results

5.1. Which denominator supports which claim

Table 2 summarises the denominators behind every figure in this section. Comparisons are valid only within a matched denominator; the review's head-

Table 2: Denominators used in this review (full-text records unless stated).

Subset	n
Retained corpus (all records)	100
Full-text records	83
Abstract-only records	17
ML-used studies	70
ML-for-detection studies	55
Non-review studies	49
Security-primary, non-review studies	41
Security-primary, non-review, detection-ML	29
Strict-grid-scope records (Section 5.5)	79
Journal articles (full text)	62

line comparison (defensive ML use against AI-threat engagement) is reported on all of rows 4–8 so that no reading depends on which subset is chosen.

5.2. Corpus profile (RQ1, descriptive)

The corpus is 100 studies: 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 4 reports, and 3 book chapters. Publication is recent and accelerating: 1 study from 2020, 5 from 2021, 2 from 2022, 8 from 2023, 8 from 2024, 31 from 2025, and 45 from 2026; 76% of the corpus appeared in 2025 or 2026 (Figure 2a). That concentration is partly a selection effect of the recency-weighted ranking, as Figure 2's caption states, and must not be read as a finding about the literature's growth. The leading publishers are MDPI (17), Elsevier (12) and IEEE (8). Table 3 gives the profile.

5.3. How the twin is positioned (RQ1)

Across all 100 records, the adjudicated stance distribution is defensive 42, review 36, bridge 10, enabling 10, adversarial 2; security is the primary subject in 83 of 100 (Figure 3a). The distribution differs from the screening seed—which had adversarial 11, review 23—confirming that keyword-level stance labelling is unreliable and that reading is necessary (Figure 3b).

The sharper result is `twin_role`, coded on the 83 full-text records: the twin is treated as a defensive *tool* in 75, as an attack *target* in exactly 1, as *both* in 0, and is *unclear* in 7. The single strict *target* record [11] is a vehicle-twin threat-modelling review, outside the grid domain; under a strict grid scope the target count is zero. The *both* = 0 result is definitional

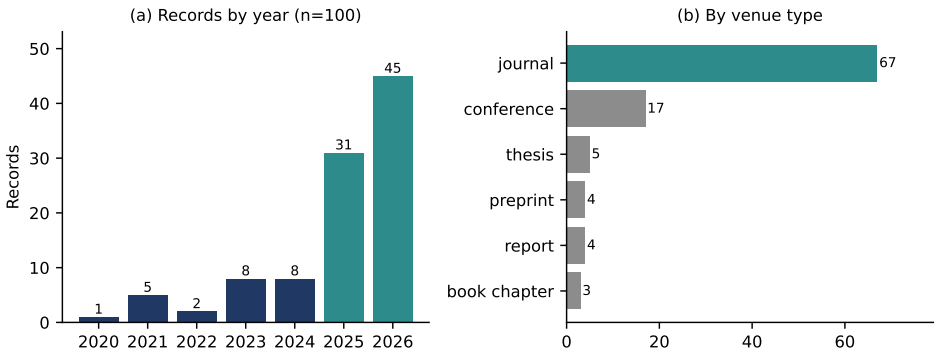


Figure 2: Corpus profile. (a) Records by year. The 2025–2026 concentration reflects selection as much as literature growth, since the top-100 cut favours recency among tied relevance scores. (b) Records by venue type.

(the codebook requires explicit articulation of the tension), so it records the absence of *explicit articulation* of the dual role that Habeeb et al. name [7], not the absence of dual-role practice.

Because "asset under protection" and "adversary's objective" are not separable in a security setting, Table 4 reports the tiered measurement side by side. Beyond the single strict-target record, 56 of 83 full-text studies name an attack on the twin's data path (man-in-the-middle 26, replay 23, reverse pathway 10, false command injection 22; interface threats combined 44) or on its learned components (poisoning, evasion, prompt-level; 27, overlapping the interface set in 15 records). Of the 75 *tool* records, 51 name such an attack channel: the dominant pattern in the corpus is a twin defended against attacks whose target surface is the twin itself, without the twin being treated as the adversary's objective. The most direct field for this claim, `attack_surface_named`, has prevalence 55/83 (66.3%) and the lowest agreement in the reliability study ($\kappa = 0.14$, PABAK 0.20; Section 5.7), so the tier boundaries carry that uncertainty.

This is the first half of the asymmetry, stated precisely: *no grid study in this corpus treats the twin as the adversary's objective*, while the twin's data path and learned components are treated as attack surfaces—to be defended—in roughly two thirds of the corpus.

5.4. Threat vocabulary and lifecycle concentration (RQ2)

Table 5 reports the closed threat vocabulary. Denial of service (49), false data injection (46) and malware (34) lead; the AI-relevant terms that the cor-

Table 3: Corpus profile ($n = 100$; full text = 83, abstract-only = 17).

Characteristic	Records
Journal article	67
Conference paper	17
Thesis	5
Preprint	4
Report	4
Book chapter	3
2026	45
2025	31
2024	8
2023	8
2022	2
2021	5
2020	1
Stance: defensive	42
Stance: review	36
Stance: bridge	10
Stance: enabling	10
Stance: adversarial	2

pus does contain— poisoning (22), evasion (11)—are present but well behind the classical attacks. A mean of 3.77 distinct threats is named per full-text study (median 3, max 11), and 10 of 83 studies name no threat at all, which is itself notable in a security review. These counts sit on the least reliable data in the review: in the double-coded sample, `threat_models_named` exact-set agreement is 5% (mean Jaccard 0.43) and `threat_count` differs by a mean of 1.8 threats against a mean of 3.77 (Section 5.7). The vocabulary counts should therefore be read as indicative of scale and ordering, not as exact prevalences.

Threat coverage is concentrated almost entirely at the operation phase. Figure 4 crosses the threat vocabulary with `lifecycle_stage`. The design and commissioning columns are effectively empty—one design-phase study [12] and none at commissioning—while the operation column is dense. Two features of the figure need saying. First, the single design-phase record [12] names only an out-of-vocabulary threat (coded `other`), so no vocabulary row

Table 4: Tiered measurement of the twin's position relative to the attack (full text, $n = 83$). "Attack channel" means the record names at least one interface threat (`mitm`, `replay`, `reverse_pathway`, `false_command`; 44 records) or model threat (`poisoning`, `evasion`, `prompt`; 27 records; 15 records name both kinds).

Tier	Records
Twin as adversary's objective (strict `target`)	1
of which grid-domain	0
Twin's data path or models named as attack channel	56
Twin as protected asset (`tool`)	75
of which also name an attack channel	51
Twin position unclear	7
`attack_surface_named` = yes ($\kappa = 0.14$)	55

registers in the design column; the column is empty because of what the one design study names, not because it names nothing. Second, the maintenance and decommissioning columns are empty as well, and no record is coded at either stage: the codebook permits both values, so the emptiness is a finding of the coding, on a field whose agreement is the lowest interpretable in the study ($\kappa = 0.32$, PABAK 0.59). Across the corpus, lifecycle is operation in 66 of 83 studies (79.5%), *multi* in 12, unclear in 4, design in 1, and commissioning in 0, as coded. Read as single labels, security work concentrates where the twin runs at deployment, not where it is built.

Threat models are also thin. An explicit, stated threat model appears in 19 of 83 studies, and an adaptive adversary is modelled in 10—so the corpus mostly names threats without committing to an attacker model, which is precisely the weakness that makes attack-side claims hard to test.

5.5. Engagement with AI-enabled threats (RQ3)

Note on terminology: the threat-vocabulary count (`poisoning`) and the AI-threat field (`ai_poisoning`) track different constructs; `poisoning` (22 records) records any discussion of poisoning as a grid attack, while `ai_poisoning` (19) records engagement with AI-specific poisoning attacks. This is the headline, reported with matched denominators. Machine learning is used in 70 of the 83 full-text studies; its dominant role is detection (55), followed by forecasting (12), none (13), and control (3). Forecasting and control are not security functions, so the security-relevant defensive figure is 55/83 (66%),

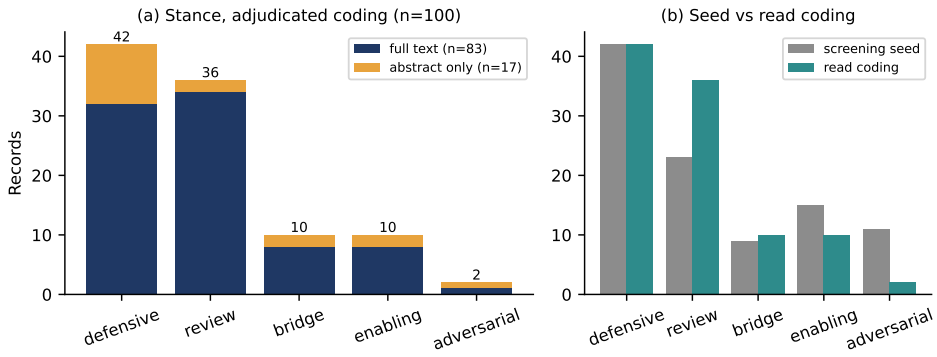


Figure 3: Stance. (a) Adjudicated coding across all 100 records, split by whether full text was coded. (b) Screening seed versus read coding: the shift from *adversarial* to *review*/*defensive* shows the seed over-called an adversarial stance on keyword evidence.

not 70/83. Across the same 83 records, 35 (42.2%) engage at least one AI-enabled threat at any level, 11 (13.3%) engage one substantively, and 48 (57.8%) do not engage at all (Figure 5). All 35 engaged studies are among the 70 that use machine learning at all, and 30 of the 55 detection-ML studies engage an AI threat, so the non-engagement among ML-using studies is 35 of 70 (50%) rather than a zero/one split. The asymmetry is real, but it is an asymmetry of emphasis rather than absence.

Three denominator restrictions probe it. Restricting to the 49 non-review studies: 20 of 49 engage any, 6 of 49 substantively. Restricting to the 41 security-primary, non-review studies: 16 any, 6 substantive. Restricting to the 29 security-primary, non-review, detection-ML studies—the subset with the strongest obligation to model attacks on their own detectors—16 of 29 (55%) engage any AI-enabled threat and 6 of 29 (21%) substantively. On every matched denominator, majority defensive use coexists with minority engagement, and substantive engagement stays at or below one fifth.

Because the corpus is a recency-weighted ranked subset, a year-wise comparison cannot support a rate claim; Table 6 reports it as description only. Within the corpus, ML use is near-universal from 2022 onward while substantive engagement never exceeds 22% in any year; the 2020–21 and 2023 cells show zero engagement on small denominators. We draw no temporal conclusion from these cells.

Per vector, at any/substantive level: adversarial machine learning 24/8, data or model poisoning 19/7, evasion 9/1, LLM-specific 4/2, and agentic

Table 5: Threat vocabulary (full text, $n = 83$). Terms are non-exclusive.

Threat	Records
Denial of service (`dos`)	49
False data injection (`fdi`)	46
Malware (`malware`)	34
Man-in-the-middle (`mitm`)	26
Replay (`replay`)	23
False command injection (`false_command`)	22
Data/model poisoning (`poisoning`)	22
Physical attack (`physical`)	21
Insider (`insider`)	20
Supply chain (`supply_chain`)	19
Model evasion (`evasion`)	11
Reverse pathway (`reverse_pathway`)	10
Other	8
Hardware	1
Prompt-level	1
No threat named at all	10
Mean distinct threats per study	3.77

attack automation 2/1.

The construct behind these counts combines two directions that should be reported separately: attacks *on* AI components and attacks *wielded with* AI. The coding does not separate them per record. A notes-based reading of the eleven substantive studies classifies four as attack generation with AI (adversarial-ML or GAN synthesis of attacks [9, 13–15]), five as attacks on learned components (poisoning, evasion or prompt-level risks [3, 5, 6, 8, 16]), and two as both [4, 17]; the any-engagement count of 35 mixes the two directions in unknown proportions. The headline is therefore an engagement figure, not an attack-type figure.

A strict-scope sensitivity addresses the breadth of the grid term. Excluding the four records outside the power-grid domain on a strict reading (records 18, 59, 69, 97: a network-traffic anomaly-detection paper with no grid or twin instantiation, the vehicle-twin review, the cyber-fraud study, and the nuclear physical-protection risk analysis) leaves 79 full-text records: 33 (41.8%) engage any AI-enabled threat, 10 (12.7%) substantively, ML use 67,

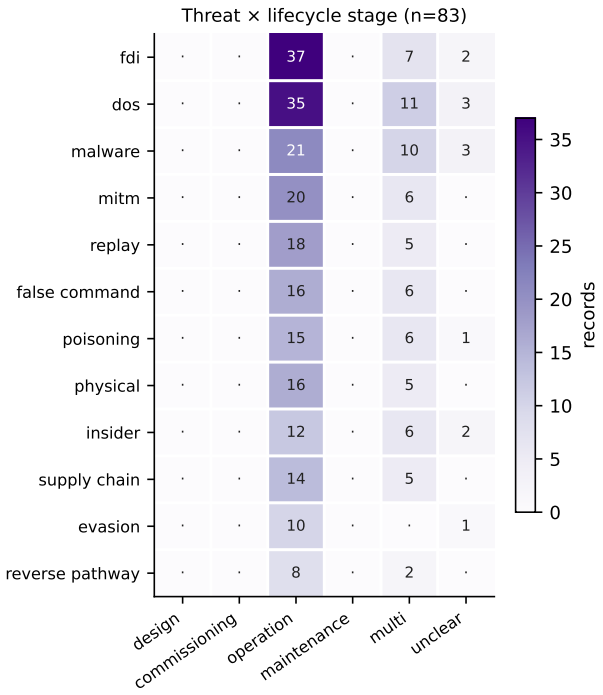


Figure 4: Threat × lifecycle-stage coverage (full text, $n = 83$). Empty cells in the design and commissioning columns are a coding outcome, not a census of practice; the field's agreement is $\kappa = 0.32$ (PABAK 0.59). The single design-phase record names only an out-of-vocabulary threat (`other`), so no vocabulary row registers in the design column. No record is coded at maintenance or decommissioning.

detection 52, and the strict twin-as-target count is zero. Excluding additionally the three borderline records (34, nuclear digital twins; 84, battery twins for telecom supply; 68, a security-education cyber twin) leaves 76 records: 32 (42.1%) any, 10 (13.2%) substantive. The asymmetry's direction is robust to scope; its magnitude moves by at most one and a half points.

Two patterns sharpen the result. First, engagement is shallow: mentioned (score 1) is more common than substantive (score 2) for every engaged vector except LLM threats (2 mentioned, 2 substantive, a tie on a base of 4) and agentic threats (1 mentioned, 1 substantive, a tie on a base of 2). Second, engagement is concentrated: every substantive evasion case, every substantive agentic case, and half the substantive LLM cases (2 of 4) sit in a single review [4], with the other LLM cases in [3]. On the review authorship question the substantive set has 5 reviews (45%) against 34 (41%) in the full-text base;

Table 6: Year-wise profile of full-text records ($n = 83$). ML = `ml_used` = yes; "any" = engages at least one AI-enabled threat; "subst." = substantive. Corpus years are shaped by the recency-weighted selection rule, so these are descriptive counts, not a temporal estimate.

Year	n	ML	any	subst.
2020	1	0	0	0
2021	3	0	0	0
2022	2	2	1	1
2023	6	4	0	0
2024	7	6	1	0
2025	27	23	15	6
2026	37	35	18	4

both counts are small and the difference is unsupported by any test.

The abstract-level screening figure is worth stating against this. Over a supplementary 469-record probe pool, an AI-enabled threat appears in the abstract in 24 records (5.1%), while the full-text corpus reads 35 of 83 (42.2%). The two figures are not like-for-like; they are quoted together as an illustration of how much screening losses can differ, not as a measured correction factor.

Table 7 lists the eleven substantive studies, with reference keys so they can be located in the bibliography. Nine are journal articles (all 2025–26), one is a thesis [8] and one a preprint [17] (2022); five carry the *review* stance and three the *bridge* stance, so 8 of 11 are review-or-bridge work. All eleven use ML predominantly for detection. No study's primary contribution is an AI-enabled attack on a grid digital twin; the deepest engagements are themselves reviews [3, 4]. One of the eleven (record 69) lies outside the grid domain on a strict reading and is excluded from the strict-scope counts above.

Read qualitatively, the eleven cluster into four patterns. (1) Generative and adversarial modelling as attack generation: the twin is used to synthesise adversarially consistent attacks and GAN-based evasion by the authors themselves [3–5, 9, 14, 17]. (2) Learned intrusion detectors stressed under distribution shift: data poisoning [8, 16, 17], FGSM/JSMA perturbations [4], and GAN-driven evasion [14, 17]. (3) LLM threats treated as architecture-level hazards—hallucination, prompt injection, biased recommendations—in the review literature [3–5]. (4) Agentic automation represented only once, in [4]; no study constructs an adversarial RL or agentic pipeline. "Substan-

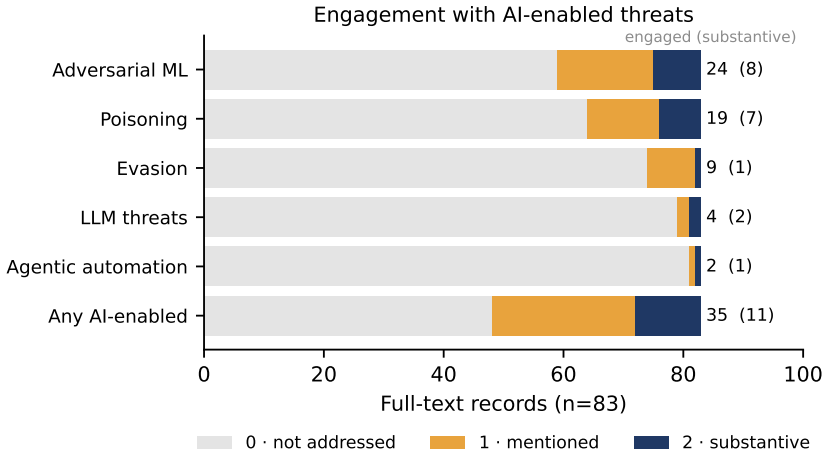


Figure 5: Engagement with AI-enabled threats (full text, $n = 83$). Scale: 0 not addressed, 1 mentioned or peripheral, 2 substantive. The bar labels give engaged (substantive) counts.

tive engage- ment" therefore currently means backgrounding a threat while retaining a defensive detector.

5.6. Evidence quality (RQ4)

Machine learning is used in 70 of 83 full-text studies; its dominant role is detection (55), followed by forecasting (12), none (13), and control (3). Evaluation method splits as simulation 33, qualitative 28, testbed 10, conceptual 5, mixed 5, and hardware-in-the-loop 2 ([18, 19] after adjudication). Maturity is prototype in 46, concept in 34, and validated in only 3. Baselines are reported in 41 of 83 studies and quantitative metrics in 43; both figures mix primary studies with reviews, so the non-review denominators matter and are reported with them: baselines rise to 37 of 49 non-review studies (76%) and metrics to 40 of 49 (82%), against 41/83 (49%) and 43/83 (52%) over all full-text records. Limitations are stated in 67. Code is released in 3 of 83 studies and data in 16. The evidence base is, in short, prototyping-heavy and release-light. A venue-quality sensitivity (restricting to the 62 journal articles) leaves the asymmetry intact (26/62 any; 9/62 substantive; 36/62 none; 58/62 tool; twin targeted by one), though it discards roughly a quarter of the corpus. Code and data remain low in every denominator.

Grid domains are dominated by studies that span multiple domains (*multi*, 47). Where a single domain is named: distribution 11, unclear 7, EV charg-

Table 7: The eleven studies engaging an AI-enabled threat substantively (score 2). IDs are corpus record numbers and double as bibliography keys (`ref<ID>`); vectors are non-exclusive. Record 69 lies outside the strict grid scope. Qualitative pattern markers are in the prose above.

ID	Ref.	Venue (type)	Year	Stance	AI-threat vectors
16	[3]	Energies (journal)	2026	review	adv. ML, poisoning, LLM
17	[16]	Sustain. Energy Technol. Assess. (journal)	2026	defensive	poisoning
22	[13]	Informatics and Automation (journal)	2026	bridge	adv. ML
24	[5]	Electronics (journal)	2026	review	adv. ML, poisoning
49	[14]	Sensors (journal)	2025	bridge	adv. ML
50	[4]	Energy Convers. Manag. X (journal)	2025	review	all five vectors
54	[6]	Energies (journal)	2025	defensive	adv. ML
57	[9]	Energies (journal)	2025	review	poisoning
69	[15]	Int. Res. J. Modernization (journal)	2025	bridge	adv. ML
74	[8]	Deep Blue, Univ. Michigan (thesis)	2025	defensive	poisoning
94	[17]	arXiv (preprint)	2022	review	adv. ML, poisoning

ing 6, microgrid 6, transmission 5, and substation 1. Among the 29 records naming a single domain, transmission (5) trails distribution (11) and EV charging (6); with 47 of 83 records coded *multi* and the field at moderate agreement ($\kappa = 0.44$), domain-level comparisons carry wide uncertainty, and we state the under-representation of transmission as an observation within this corpus rather than a claim about the literature.

5.7. Reliability results

Table 8 reports agreement on the 20-record sample (24% of the 83 full-text records). On `ai_threat_overall`, the 3x3 counts are (rows coder 1, columns coder 2): 0: 9, 1, 1; 1: 1, 3, 1; 2: 1, 2, 1. The count behind the headline, 35/83, rests on the "any engagement vs. none" collapse: coder 1 has 9 of 20, coder 2 has 9 of 20, agreement 16/20 ($\kappa = 0.60$, PABAK 0.60, bootstrap 95% CI 0.20–0.90). The three-class κ is 0.41 (PABAK 0.48)—these are different statistics for different questions, and earlier drafts conflated them. Collapsing to "substantive vs. not": agreement 15/20, but $\kappa = 0.14$ (PABAK 0.50, bootstrap 95% CI -0.23 to 0.64), because only 1 of the 6 studies either coder labelled substantive is labelled so by both (the single (2,2) cell). The two collapses are not comparably reliable.

The disagreement on "any engagement" is symmetric (2 each way), but

four disagreements cannot distinguish coder bias from noise, and symmetry is not evidence of accuracy: the interval on the difference includes under- and over-counting of the same magnitude. The headline supported by the reliability data is the 35/83 any, not the 11/83 substantive, which is fragile. This paragraph is a post-hoc reading of the 3x3 once we saw it, and is flagged as such.

To quantify what the misclassification does to the headline, we propagate the sample error matrix to the corpus under the assumption that coder 2's coding is the gold standard: the coder-1-engaged records stay engaged with probability 7/9, and the coder-1-zero records flip to engaged with probability 2/11, giving an expected $35 \times 7/9 + 48 \times 2/11 \approx 36$ of 83 (43.3%; bootstrap 95% CI 21–51). Coder 2 is a co-author of this review, so this is a consistency check, not an independent validation; it says the headline does not move materially under the observed error structure, and that the uncertainty band around 35/83 is roughly ±15 records.

Fields with substantial agreement include `baseline_compared` ($\kappa = 0.90$), `twin_role` (0.77), `ai_poisoning` (0.73), `threat_model_explicit` (0.69) and `metrics_reported` (0.68)—though `twin_role`'s κ carries a bootstrap interval of 0.00–1.00 and the sample contains about 0.2 expected *target* cases, so it says almost nothing about the rare class the headline needs. Fields at moderate or fair agreement include the ones the headline most needs: `ai_threat_overall` (0.41), `ai_evasion` (0.39), and the threat-vocabulary fields, which are the least reliable data in the review (exact-set agreement 5%). `attack_surface_named` is the weakest interpretable field (0.14, PABAK 0.20). Where a field is near-constant, κ is undefined and we report percentage agreement instead: `stance_confidence`, `limitations_stated`, `data_available`, `ai_llm`, `ai_agentic` and `code_available`.

We report these numbers because they bound the result. The headline's core count—whether a study engages any AI-enabled threat—is coded at moderate agreement (0.60 on the collapse that produces it), so the 35/83 figure carries real uncertainty. What the design can and cannot support should be stated plainly: adjudication cannot test the headline (its tie-break defaults to coder 1), and the false-negative audit shares the extraction's keyword blind spot (Section 5.8); the reliability statistics and the misclassification sensitivity above are the only quantitative handles on the headline's uncertainty, and both come from a sample of 20 coded by two authors of this review. The asymmetry's *direction* is supported on every denominator and scope re-

striction we report; its *exact magnitude* should be quoted with the interval above.

A systematic style difference between coders emerged, and the bias indices in Table 8 quantify it rather than assert it: the largest between-coder marginal differences are on `maturity` (0.35), `stance` (0.30) and `attack_surface_named` (0.30), with coder 1 assigning the default category (*high*, *yes*, *operation*, *single*) and coder 2 differentiating (*medium*, *no*, *multi*, *unclear*) and naming more attack surfaces. PABAK, which removes prevalence depression, lifts every field's reading—most sharply the near-constant ones—so where κ and PABAK disagree, the truth is that the coders agreed often but the disagreements ran one way. We record this as a codebook problem rather than a coder problem, and as a reason not to quote κ alone on this sample.

5.8. False-negative audit

The strict AI-threat lexicon flagged none of the 48 records coded as not engaging, which is consistent with the coding—but, as the methods state, the audit shares the extraction's keyword blind spot, so this "zero" is weak evidence by construction. As a positive control the same lexicon recovers only 7 of 11 known-positive studies, because some positives use generic vocabulary the lexicon lacks (e.g. [9, 13, 15, 16]). Recall on score-1 (mentioned-only) engagement, which covers 24 of the 35 engaged records and is the harder case, was never measured: no score-1 positive control was run. A pattern set cannot recognise wording it was not written for, and this audit is a pattern set applied to the output of a pattern-assisted extraction.

We therefore also ran a broader eleven-family net. Of the 48 negatives, 38 fired on some family and 23 fired on a strong family. All 23 strong firings were individually inspected and are either conventional cyberattacks discussed without any AI component or known false positives. The instructive false positives justify human coding over keyword coding: `GaN` (the semiconductor) matching "GAN" in two records [10, 20], the author surname "Gan" in another [21], Xenon-135 reactor physics matching "poisoning" [22], ARP poisoning [23], and the SolarWinds incident discussed as a data-poisoning example [24]. Two close calls were examined and cleared: [25] bypasses *classical* state-estimation bad-data detection, not an ML detector, and [26] concerns a SCADA supervisor with AI detection stated only as future work. The remaining 15 weak firings were *not* individually inspected, and are reported here as uninspected; their unexamined status is part of the audit's limit, not an oversight we claim to have covered. No codes were changed.

What this audit licenses is therefore narrow: the false-negative channel is disclosed, not bounded. The design that would bound it is an independent full-text re-read of a random sample of the 48 negatives by a coder outside the author team, with a miss-rate estimate and confidence interval propagated to the 35/83 figure. That re-read is listed as required future work in Section 8; it cannot be reconstructed from the released keyword digests alone, because the full texts are not redistributable with the package.

5.9. Adjudication outcome

Adjudication resolved 149 field-level disagreements and changed 20 net cells from coder 1's values: `ml_family` (12), `threat_models_named` (2), `stance` (2), `eval_method` (2), `dt_type` (1) and `grid_domain` (1). Three substantive corrections came out of it: two studies move to *bridge* stance [13, 14], and one study is a substation-domain study [19], contrary to its initial coding. Two third-value resolutions occurred, both `eval_method` = hardware-in-the-loop [18, 19]. No headline field moved. Under the coder-1-default tie-break this is the expected outcome (Section 4), so it should not be read as evidence that the headline is robust to adjudication; the honest statement is that adjudication, in this design, is not a test of the headline at all.

6. Discussion

6.1. The asymmetry, quantified

The corpus uses machine learning in 70 of 83 full-text studies—for detection in 55—and engages AI-enabled threats in 35, substantively in 11; within the studies most obliged to model attacks on their own detectors (security-primary, non-review, detection-ML, $n = 29$) the figures are 16 any and 6 substantive. That gap—between majority defensive use and minority, mostly peripheral attack-side engagement—is the paper's central finding. Its robustness rests on the denominator and scope restrictions of Section 5.5 and on the misclassification sensitivity of Section 5.7, not on the adjudication or the keyword audit, neither of which can test it (Sections 5.8 and 5.7). The practical reading is that the field has built a large defensive apparatus on models whose own failure modes it rarely studies. We state the finding as an asymmetry of emphasis, not of rate: no temporal claim is supported by a recency-weighted ranked corpus.

6.2. The twin as a target: strict count zero, loose tiers two-thirds

No grid study in this corpus treats the twin as the adversary's objective; the single strict *target* record [11] concerns vehicle twins and falls outside the grid domain. Habeeb et al. name the twin as an expanded attack surface [7]; on the loose tiers of Table 4, 56 of 83 full-text studies name an attack channel on the twin's data path or learned components, and 51 of the 75 *tool* records do so while treating the twin as an asset under protection. The honest summary is therefore not that attack-surface engagement is absent, but that it is *defensive almost everywhere*: the twin's surfaces are named as things to protect, and in no grid study is the twin itself the object of the attack. What we cannot speak to is the broader version of their axis—studies in which the twin's expanded surface is the analytical centre—because our coding separates objective from asset, not centre from periphery.

6.3. The lifecycle concentration, read carefully

A descriptive result with real reach, if it holds, is the lifecycle concentration: 66 of 83 studies are coded at the operation phase, one at design, none at commissioning or maintenance. Two caveats apply. First, the field carries our lowest interpretable agreement ($\kappa = 0.32$, PABAK 0.59), and coder 1, who coded 80% of the corpus, defaults to 'operation'; the 12 studies coded 'multi' may contain design or commissioning activity that the single label suppresses. Second, the score records the phase a study *targets*, not the phases it necessarily *touches*. Read conservatively, the finding is therefore that the literature is dominated by studies of running twins, with almost no study that sets out to secure a twin before it is built, not that no design-phase security work exists. We present this as a descriptive observation on the weakest-reliability field in the review, not as a headline result.

6.4. Why reviews are where the threats are named

The two deepest engagements are reviews [3, 4]; 5 of the 11 substantive studies (45%) are reviews against 34 of 83 (41%) in the full-text corpus, so review authorship is not over-represented, but bridge studies are (3 of 11 against 8 of 83 full-text records, 2 more in the abstract-only set). The honest reading is that reviews have the length and the licence to survey AI threats, while primary papers have neither. Document length is a clear confound and these numbers should not be read as evidence that 'only reviewers care.' Also, the abstract-versus-full-text comparison in Section 5.8 is not like-for-like (24/469 comes from a wider query set and a different threshold than

35/83) and should be read as an illustration of screening loss rather than an 8x correction factor.

6.5. Positioning against the prior review

Relative to Habeeb et al. [7], our contribution is a grid-side probe of a narrower instance of one of their axes, with a published corpus, a closed vocabulary, and reported reliability; full scope alignment is in Section 6.2. Table 1 shows that our independent vocabulary covers all five of their vectors, so the comparison is not competing but additive. Two caveats are structural, not rhetorical. First, their search string has no specific attack-side operator, so records enter only through security terms; the same is true of our twelve core queries, and our four adversarial-AI probes were added precisely because of that shared blind spot. We can no longer claim the probes as a completed defence, however: the released record does not establish that probe-only records entered the retained corpus (Section 4), so the shallow engagement we measure may be partly a retrieval artefact of the core vocabulary, and the asymmetry could be weaker in the un-retrieved attack-side literature than in this corpus. Resolving that requires releasing per-record query provenance and re-running the headline on the probe-only subset—the concrete step recorded in the repository's provenance manifest. Second, their corpus spans manufacturing, healthcare, water and smart city, where the digital twin's role differs from the grid's bidirectional, safety-critical loop; grid conclusions should not be generalised from it, nor it from us.

6.6. Operational reading

For operators, three interpretive notes follow (these are author readings, not coded evidence). First, a twin whose learner is trained on the same telemetry it is meant to defend has no independent ground truth and is a single point of failure that the corpus rarely names. Second, runtime detection at the operation phase is necessary, but cannot cover design-stage trust failures. Third, the near-total absence of released code and data means a twin procurement decision cannot currently be informed by reproducible attack evidence.

7. Research agenda

We state five items. Each is tied to the gap that justifies it (from Section 5), to the validation that would confirm it, and to the evidence that

would *disconfirm* it, so the agenda is falsifiable rather than aspirational. Table 9 summarises.

P1 — Attack grid twins as twins, and study the twin's surface in its own right. Justified by the tiered result: strict `target` = 0 grid studies against 56/83 naming an attack channel on the twin's data path or models (Table 4); confirmed by a reproducible attack against a grid twin in which the twin (not the plant) is the objective, with released code and a stated adaptive adversary; disconfirmed by primary studies that already construct such attacks (which would move the strict count and end this gap).

P2 — Measure lifecycle coverage, then extend it to design and commissioning. Justified by operation = 66, design = 1, commissioning = 0, on a field with $\kappa = 0.32$ (PABAK 0.59); confirmed by twins whose trust boundaries, interface contracts and fail-safe behaviour are specified and tested before deployment, and by a double-coded re-measurement that survives the lifecycle field's reliability problem; disconfirmed by a re-measurement showing the concentration to be a coding artefact of the single-label default.

P3 — Adopt explicit adaptive threat models for the twin's learned components. Justified by an explicit threat model in 19/83 and an adaptive adversary in 10/83 against detection-ML use in 55/83; confirmed by studies that state the attacker's knowledge, capability and objective *relative to the twin's detectors and forecasters* (e.g. an attacker who retrains against the twin's anomaly detector or probes its feedback loop) and test a reactive attacker; disconfirmed by primary studies that already state and test such models, which the corpus currently lacks.

P4 — Defend the learned components, not just the plant. Justified by detection-dominant ML use (55/83) against shallow poisoning and evasion engagement (19 and 9 at any level, 7 and 1 substantive); confirmed by poisoning- and evasion-robust twins under distribution shift; disconfirmed by primary studies demonstrating robust learned detectors whose robustness is evaluated against a reactive attacker.

P5 — Release code, data and attack generators. Justified by code in 3/83 (3/49 non-review) and data in 16/83 against a corpus whose

attack-side claims cannot be re-run by third parties; confirmed by a third party rerunning a published twin's attack evaluation unaided; disconfirmed by the corpus remaining release-light while attack-side claims multiply. This item is AI-specific in its object: what must be released are the learned models, their training data, and the attack generators evaluated against them—artifacts whose absence makes adversarial claims unverifiable in a way that releasing only simulation scripts would not fix.

P5 must be stated carefully. We measure *research attention*, not *deployment security*. A low attack-side count does not mean grids are undefended; it means the published research has not modelled these attacks. We cannot estimate either the probability of an AI-enabled attack or its expected impact from this corpus, and we do not.

8. Threats to validity

8.1. Construct validity

"AI-enabled threat" is a construct with two distinct directions lumped together: attacks *on* AI components and attacks *wielded with* AI. The coding does not separate them per record, so the headline 35/83 is an engagement figure, not an attack-type figure (Section 5.5 reports the notes-based split for the substantive studies). The false-negative audit shows a strict lexicon misses some positives (7/11 recovery), and shares the extraction's keyword blind spot, so it discloses rather than bounds the false-negative channel. We mitigated by human coding with context, by the broad eleven-family net, and by the tiered `twin_role` reporting, but the construct remains softer than the threat-vocabulary fields.

8.2. Internal validity and screening

Screening on title and abstract is imperfect: in the probes built into this study, an AI-threat term appears in the abstract in 24 of 469 (5.1%) records, whereas the retained full-text corpus reads 35 of 83 (42.2%). The two figures use different query sets and different screening thresholds, so they are an illustration of screening loss, not a like-for-like comparison, and the 'order of magnitude' distance between them should not be quoted.

8.3. Reliability

Agreement on the headline fields is moderate, not strong: the any-engagement collapse that produces 35/83 is $\kappa = 0.60$ (bootstrap 95% CI 0.20–0.90; the three-class κ is 0.41); `attack_surface_named` is 0.14 (PABAK 0.20). Both coders are authors of this review, the sample is 20 of 83 full-text records, and coder-1 precedence in adjudication is a declared limitation that makes adjudication uninformative about the headline. The exact magnitude of the asymmetry should therefore be read with the misclassification sensitivity (36 of 83, 95% CI 21–51), not in isolation.

8.4. External validity and coverage

This is not a PRISMA-systematic review: one database, no stated search date in the text, no registered protocol, a ranked (not random) retained subset, and five codebook revisions after the protocol was 'fixed'—all of which limit how strongly the review answers its hypothesis. Because the corpus is a ranked subset, no prevalence figure here estimates the wider literature, and no sampling-theory interval (Wilson or otherwise) is reported; the misclassification sensitivity of Section 5.7 is the only interval we attach to the headline.

Seventeen of the 100 records were available only as abstracts; this is an access limitation of the authors, not a property of the literature, and those records are listed with DOIs (Appendix 9) so readers can check what is missing. One corpus DOI does not resolve (record 18, which lies within the reliability sample) and three records have no DOI; these are disclosed rather than dropped. Venue labels disagreed with Crossref in 18 of 54 cases and were corrected in favour of the registry. The broad grid term admits adjacent domains; the strict-scope sensitivity (Section 5.5) shows the headline moves by at most one and a half percentage points under the strictest reading.

8.5. Search bias

The core grid-security vocabulary does not retrieve attack-side work; the four adversarial-AI probes were added to widen the candidate pool (approximately 2,000 records), but the released record does not establish that probe-only records entered the screened set or the final 100, and the query strings and per-record provenance were not preserved in the released package. We therefore withdraw the earlier claim that the probes make the asymmetry robust to retrieval bias: on the released evidence, the corpus derives from

the core grid-security vocabulary alone, which is precisely the retrieval bias the probes were meant to fix, and the asymmetry may be overstated in the un-retrieved attack-side literature. The repository's provenance manifest ('SEARCH_PROVENANCE.md') states the artifacts whose release would resolve this (query strings, run dates, per-record provenance, and a re-run of the headline on the probe-only subset); until then, the corpus should be read as a bounded sample, not a census. The recency component of the ranking also means foundational attack-side work from 2020–2023 may be under-weighted, and the citation component disadvantages new attack-side work that has not yet accumulated citations.

8.6. Researcher degrees of freedom

Five codebook revisions during extraction are a real degree of freedom. They are logged, dated, and applied consistently to all 100 records, and adjudication used the same rules; but a reader who distrusts post-hoc refinement should weight the raw pre-adjudication codes, which are released frozen alongside the adjudicated matrix.

8.7. Quality appraisal, venue quality and scope

At least one retained record [27] carries anonymous authorship in a low-tier journal, which is exactly why the venue sensitivity and DOI verification passes above were necessary. The corpus contains 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 4 reports and 3 book chapters, drawn from venues of varying strictness (including IRJMETS and GSC Online Press). No formal risk-of-bias or venue-quality appraisal was performed; all prevalence figures are therefore reported over the full corpus, and Section 5 reports sensitivity subsets for non-review studies. Readers should weight venue quality into any claim reproduced from this review. The claim in Section 1 is restricted to the intersection of 'digital twin' and 'cybersecurity' in power grids; substantial adversarial-ML work on grid state estimation lies outside that intersection and is not covered by the headline numbers.

9. Conclusion

In this corpus, grid digital-twin security research uses machine learning for defence far more often than it engages AI-enabled threats or the twin itself as an attack target. Across a ranked corpus of 100 studies (a declared

non-probabilistic subset of 492 eligible records, so every figure describes the corpus, not the literature; its 76% share of 2025–26 records is partly a selection artefact), machine learning is used in 70 of 83 full-text studies—for detection in 55—while 35 engage any AI-enabled threat, 11 substantively, and 16 of the 29 security-primary, non-review, detection-ML studies engage any. No grid study treats the twin as the adversary's objective; 56 of 83 name an attack channel on the twin's data path or learned components, almost always as something to protect. Coverage, as coded, is dominated by the operation phase (66 of 83, on the field with the lowest interpretable agreement, $\kappa = 0.32$); one study is coded at design and none at commissioning or maintenance, and threat models are thin (explicit in 19/83).

These directions are robust to the denominator and scope restrictions we report (including a strict grid scope) and to a misclassification sensitivity anchored on the second coder (36 of 83, 95% CI 21–51). They are not tested by the adjudication, whose tie-break defaults to the primary coder, nor bounded by the false-negative audit, which shares the keyword extraction's blind spot; both limits are stated as such above. The closest prior review names the dual role; this review measures only the narrowest, strict-objective instance of its attack-surface axis at grid scale and finds that instance empty, while the looser tiers show the twin's surfaces widely engaged but almost always defensively. The broader attack-surface axis remains untested here. The five-item agenda that follows is deliberately falsifiable: each item names the gap that justifies it, the evidence that would close it, and the evidence that would show the gap was never real.

Highlights

- Structured review of a ranked 100-study corpus on digital twins for power-grid cybersecurity (2020–2026); figures describe the corpus, not the literature.
- ML is used in 70/83 full-text studies (detection in 55); 35/83 engage an AI-enabled threat, 11/83 substantively (fragile, $\kappa = 0.14$); 16/29 of the security-primary, non-review, detection-ML studies engage any.
- No grid study treats the twin as the adversary's objective; 56/83 name an attack channel on the twin's data path or models, almost always as something to protect.

- Coverage, as coded: 66/83 at operation, 1 at design, 0 at commissioning (field agreement $\kappa = 0.32$).
- Five-item agenda, each item tied to its gap, its validating evidence, and the evidence that would disconfirm it.

Declaration of competing interest

The authors declare no competing interests.

Data availability

All extraction data, the codebook, the coder sheets, the reliability script and the figure script are available in the reproducibility package accompanying this review (repository link supplied at submission). Appendix 9 lists all 100 included studies in the paper itself; the search-provenance manifest (`SEARCH_PROVENANCE.md`) records the query-level artifacts that remain to be released, as Section 8 discloses.

CRediT authorship contribution statement

Author One: conceptualisation, methodology, formal analysis, writing—original draft. **Author Two:** validation, reliability coding, writing—review and editing.

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Appendix A. The 100 included studies

Table 10 lists every record in the retained corpus, so that the included-study set is visible in the paper itself and not only in the released data. IDs are corpus record numbers and double as bibliography keys (`ref<ID>`).

"FT" marks the 83 records coded at full text; the 17 abstract-only records were coded at abstract level, with full-text fields treated as missing. Records 18, 59, 69 and 97 fall outside the strict grid scope of Section 5.5; records 34, 68 and 84 are the borderline cases there.

Table 10: The retained corpus ($n = 100$).

ID	Study (first author)	Venue (type)	FT	DOI
1	Bo Nørregaard Jørgensen (2026)	Applied Sciences (journal)	Y	10.3390/app16020981
2	N Subramanian (2026)	Energy Reports (journal)	Y	10.1016/j.egyr.2026.109218
3	Muhammed Cavus (2026)	Renewable and Sustainable Energy Reviews (journal)	Y	10.1016/j.rser.2026.116947
4	Deepthi Pilakkat (2026)	IEEE Access (journal)	Y	10.1109/access.2026.3674387
5	Ehsan Naderi (2026)	Electronics (journal)	Y	10.3390/electronics15091894
6	Binastya Anggara Sekti (2026)	Advances in computational intelligence ... (chapter)	–	10.4018/979-8-3373-4159-0.ch009
7	Nagarajan Munusamy (2026)	Global Energy Interconnection (journal)	Y	10.1016/j.gloei.2026.01.002
8	Giuseppe Loseto (2026)	IEEE Sensors Journal (journal)	–	10.1109/jsen.2026.3675872
9	Saad Khan (2026)	IEEE Access (journal)	Y	10.1109/access.2026.3659724
10	Matteo Repetto (2026)	Applied Cybersecurity & Internet Gover... (journal)	Y	10.60097/acig/221493
11	Anant Upadhiyay (2026)	Discover Internet of Things (journal)	Y	10.1007/s43926-026-00386-8

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
12	Francisca Ezulike (2026)	International Conference on Cyber Warfa... (conf.)	Y	10.34190/iccws.21.1.4467
13	Ismail Erkek (2026)	International Journal of Information Se... (journal)	Y	10.1007/s10207-026-01274-6
14	Syed Saad Ali (2026)	International Journal of Innovations in... (journal)	Y	10.33411/ijist/1995
15	Roberto Ciavarella (2026)	Electricity (journal)	Y	10.3390/electricity7030085
16	Reham Abdullah Sanad Alsbua (2026)	Energies (journal)	Y	10.3390/en19153643
17	Hao Zhang (2026)	Sustainable Energy Technologies and Ass... (journal)	Y	10.1016/j.seta.2026.105246
18	Syed Mehzeyar Ali Zaidi (2026)	Journal of Engineering and Computationa... (journal)	Y	10.63544/qcr0y085
19	Shameem Hasan (2026)	Energy Conversion and Management X (journal)	Y	10.1016/j.ecmx.2026.102107
20	Aravind Kumar Karpoorapu (2026)	Zenodo (CERN European Organization for ... (chapter)	Y	10.5281/zenodo.22763194
21	Yue Wang (2026)	Systems (journal)	Y	10.3390/systems14101271
22	Igor Vitalievich Kotenko (2026)	Informatics and Automation (journal)	Y	10.15622/ia.25.3.8
23	S. Rayhan Kabir (2026)	Applied Sciences (journal)	Y	10.3390/app16157655

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
24	Osasumwen Cedric Ogiesoba-Eguakun (2026)	Electronics (journal)	Y	10.3390/electronics15204701
25	Oleg V. Trubienko (2026)	Russian Technological Journal (journal)	Y	10.32362/2500-316x-2026-14-5-41-54
26	Paroma Chatterjee (2026)	unknown-venue (thesis)	–	10.33915/etd.13429
27	Kanchan Yadav (2026)	unknown-venue (conf.)	–	10.1109/parc68365.2026.1145352
28	Jennifer Ikeoluwa Ademoomiyo (2026)	International Journal of Innovative Tec... (journal)	Y	10.35940/ijitee.k1309.15111026
29	GHOREISHI, MOHAMMAD (2026)	IRIS Research product catalog (Sapienza... (thesis)	–	—
30	Safa Mohamed (2026)	Zenodo (CERN European Organization for ... (journal)	Y	10.5281/zenodo.19999925
31	Franco Fernando Yanine (2026)	Sustainability (journal)	Y	10.3390/su18178724
32	Subiksha GD, Anusha C, Deno Star A (2026)	Zenodo (CERN European Organization for ... (journal)	Y	10.5281/zenodo.19666287
33	Syed Kousar Niasi K (2026)	Preprints.org (preprint)	Y	10.20944/preprints202601.0392.v
34	Mehtab Alam (2026)	Frontiers in Mechanical Engineering (journal)	Y	10.3389/fmech.2026.1926649
35	Fahad Khan (2026)	International Journal of Advanced Research (journal)	Y	10.21474/ijar01/24257

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
36	Vladyslav Pliuhin (2026)	Lighting Engineering & Power Engineering (journal)	Y	10.33042/2079-424x.2026.65.1.06
37	Allen Antony (2026)	Hogskolan Ihalmstad (Halmstad University) (thesis)	–	—
38	Narendra Kumar (2026)	International Journal of Digitalization (journal)	Y	10.20508/rrhh1109
39	Zornitsa B. Yordanova (2026)	Resilient Cities and Structures (journal)	Y	10.1016/j.rcns.2026.09.003
40	Akram AL-DHAFARI (2026)	unknown-venue (report)	–	10.1002/9781394474325.ch14
41	Somayeh Soroush (2026)	Electronics (journal)	Y	10.3390/electronics15194522
42	Peng Xiao (2026)	Ingegneria Sismica (journal)	–	10.65102/is2026854
43	Akriti . (2026)	International Journal of Artificial Int... (journal)	Y	10.51483/ijaiml.6.9s.2026.1975-1984
44	Kabiru Alani Hassan (2026)	Journal of Multi-Disciplinary Engineeri... (journal)	Y	—
45	Emmanuel Emenike Ezeoba (2026)	Advances in Machine Learning & Artific... (journal)	Y	10.33140/amlai.07.03.03
46	T. A. Rajaperumal (2025)	Energy Informatics (journal)	Y	10.1186/s42162-024-00461-w
47	Sohel Rana (2025)	American Journal of Advanced Technology... (journal)	Y	10.63125/4p25x993
48	Herman Zahid (2025)	Results in Engineering (journal)	Y	10.1016/j.rineng.2025.105850

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
49	Ali Sayghe (2025)	Sensors (journal)	Y	10.3390/s25164963
50	Yaser Mike Banad (2025)	Energy Conversion and Management X (journal)	Y	10.1016/j.ecmx.2025.101329
51	Hind Lakhal (2025)	IEEE Access (journal)	Y	10.1109/access.2025.3623834
52	Muhammed Cavus (2025)	Energies (journal)	Y	10.3390/en18174599
53	Reza Hemmati (2025)	Energy Conversion and Management (journal)	–	10.1016/j.enconman.2025.120728
54	Bo Li (2025)	Energies (journal)	Y	10.3390/en18123145
55	Samuel Abiodun Oyedotun (2025)	Journal of Science Research and Reviews (journal)	Y	10.70882/josrar.2025.v2i3.76
56	Kwabena Addo (2025)	Energies (journal)	Y	10.3390/en18215593
57	Ernest Fiko Morgan (2025)	Energies (journal)	Y	10.3390/en18226048
58	Nabeel A. Y. Al-Qirim (2025)	International Journal of Engineering Bu... (journal)	Y	10.1177/18479790251328183
59	Uzair Muzamil Shah (2025)	Smart Cities (journal)	Y	10.3390/smartcities8050142
60	Pradeep Chintale (2025)	Auerbach Publications eBooks (chapter)	–	10.1201/9781003500391-7
61	Evgenii Sergeevich Mityakov (2025)	Computational nanotechnology (journal)	–	10.33693/2313-223x-2025-12-4-116-123
62	Mansi Girdhar (2025)	IEEE Access (journal)	Y	10.1109/access.2025.3648339
63	He Jian Wu (2025)	Preprints.org (preprint)	Y	10.20944/preprints202506.0030.v
64	Ghiulnar - Yelda Cherim (2025)	International Symposium on Advanced Top... (conf.)	Y	10.1109/atee66006.2025.1129988

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
65	Pedro Barros (2025)	World Journal of Advanced Research and ... (journal)	Y	10.30574/wjarr.2025.26.3.2233
66	Muhammad Arbab Khan (2025)	Global Power, Energy and Communication ... (conf.)	Y	10.1109/gpecom65896.2025.1106
67	Samir Thapa (2025)	IEEE International Conference on Electr... (conf.)	Y	10.1109/eit64391.2025.11103680
68	Vita Santa Barletta (2025)	Biannual Conference of the Italian SIGC... (conf.)	Y	10.1145/3750069.3750102
69	Unattributed (2025)	International Research Journal of Moder... (journal)	Y	10.56726/irjmets81350
70	Li, Jin (2025)	Espace ETS (ETS) (report)	Y	10.82417/cc0h-fz32
71	Saif Ali (2025)	Euro vantage journal of artificial inte... (journal)	Y	10.65923/avg24p53
72	Dalila Anseri (2025)	unknown-venue (conf.)	–	10.1109/ictaacs69003.2025.11399
73	Bilal Khan (2025)	International Conference on Electrical,... (conf.)	Y	10.1109/icecce67514.2025.112580
74	Mansi Girdhar (2025)	Deep Blue (University of Michigan) (thesis)	Y	10.7302/25171
75	Artem I. Zemtsov (2025)	Gylym zane bilim (journal)	Y	10.52578/2305-9397-2025-3-5-328-336
76	Abraham Ayodeji Abayomi (2025)	International Journal of Scientific Res... (journal)	Y	10.32628/cseit251234

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
77	Opy Das (2024)	Energy Conversion and Management X (journal)	Y	10.1016/j.ecmx.2024.100715
78	Anurag Kumar Srivastava (2024)	IEEE Power and Energy Magazine (journal)	–	10.1109/mpe.2023.3325196
79	Nabeel A. Y. Al-Qirim (2024)	IEEE Open Journal of the Communications... (journal)	Y	10.1109/ojcoms.2024.3502544
80	Baza Somda (2024)	International Conference on Intelligent... (conf.)	Y	10.1109/isap63260.2024.1074437
81	Timotius Oentung (2024)	Resilience Week (RWS) (conf.)	Y	10.1109/rws62797.2024.1079944
82	Tommaso Zoppi (2024)	IEEE Pacific Rim International Symposiu... (conf.)	Y	10.1109/prdc63035.2024.00039
83	Abdulmueen Alrashide (2024)	Proceedings of the Annual Conference of... (conf.)	Y	10.1109/iecon55916.2024.109053
84	Masoud Amirrezai (2024)	IEEE Industrial Electronics Society Ann... (conf.)	Y	10.1109/oncon62778.2024.10931
85	Prabhat Kumar (2023)	Solar Energy (journal)	Y	10.1016/j.solener.2023.111921
86	Mansoor Ali (2023)	IEEE Transactions on Information Forens... (journal)	Y	10.1109/tifs.2023.3305916
87	Erdal Irmak (2023)	Energies (journal)	Y	10.3390/en16124590
88	Omer Sen (2023)	IEEE International Conference on Commun... (conf.)	Y	10.1109/smartgridcomm57358.20
89	Gholamreza Nasiri (2023)	Machines (journal)	Y	10.3390/machines11030392

Table 10 continued

ID	Study (first author)	Venue (type)	FT	DOI
90	Luka V. Strezoski (2023)	Energija Ekonomija Ekologija (journal)	Y	10.46793/eee23-4.01s
91	Christopher M. Spirito (2023)	unknown-venue (report)	–	10.2172/2337610
92	Christopher M. Spirito (2023)	unknown-venue (report)	–	10.2172/2331253
93	Md. Mhamud Hussen Sifat (2022)	Energy and AI (journal)	Y	10.1016/j.egyai.2022.100213
94	Tianming Zheng (2022)	arXiv (Cornell University) (preprint)	Y	10.48550/arxiv.2205.11783
95	William Danilczyk (2021)	North American Power Symposium (NAPS) (conf.)	–	10.1109/naps50074.2021.9449682
96	Amardeep Balasaheb Shitole (2021)	IEEE International Conference on Smart ... (conf.)	Y	10.1109/stpec52385.2021.971861
97	Yun Guo (2021)	International Conference on Power Syste... (conf.)	Y	10.1109/powercon53785.2021.969
98	Nandha Kumar Kandasamy (2021)	arXiv (Cornell University) (preprint)	Y	10.48550/arxiv.2105.04260
99	Ahmed Aly Saad Ahmed (2021)	unknown-venue (thesis)	–	10.25148/etd.fidc009671
100	Ahmed Saad (2020)	IEEE Transactions on Smart Grid (journal)	Y	10.1109/tsg.2020.3000958

Table 8: Inter-coder agreement on the blind sample ($n = 20$ of the 83 full-text records). κ is reported where interpretable; PABAK uses the codebook's category count per field; "bias" is the largest between-coder difference in category marginals (proportion of 20); † marks near-constant fields (modal category $\geq 90\%$ in coder 1) where κ is not interpretable and percentage agreement is reported instead. Bootstrap 95% CIs for headline fields are in the text.

Field	% agree	κ / set metric	PABAK	bias
Cohen's κ interpretable				
baseline_compared	95	0.90	0.90	0.05
twin_role	95	0.77	0.93	0.05
ai_poisoning	90	0.73	0.85	0.10
threat_model_explicit	90	0.69	0.80	0.10
metrics_reported	85	0.68	0.70	0.05
security_primary	85	0.58	0.70	0.15
ml_used	85	0.57	0.77	0.05
ml_role	70	0.55	0.64	0.10
stance	60	0.46	0.50	0.30
dt_type	75	0.46	0.69	0.15
grid_domain	65	0.44	0.61	0.15
ai_adv_ml	70	0.44	0.55	0.10
maturity	60	0.42	0.47	0.35
ai_threat_overall	**65**	**0.41**	**0.48**	**0.05**
ai_evasion	75	0.39	0.62	0.20
eval_method	50	0.37	0.42	0.20
lifecycle_stage	65	0.32	0.59	0.25
attack_surface_named	60	0.14	0.20	0.30
Percentage agreement only (†)				
code_available	100	†	1.00	0.00
ai_agentic	95	†	0.92	0.05
ai_llm	90	†	0.85	0.10
data_available	85	†	0.77	0.10
stance_confidence	70	†	0.55	0.20
limitations_stated	70	†	0.40	0.10
Multi-label fields (set metrics)				
threat_models_named	5	Jaccard 0.43	—	—
ml_family	20	Jaccard 0.42	—	—
threat_count	10	mean $\lvert\Delta\rvert$ 1.80	—	—

Table 9: Research agenda: gap, direction, the validation each item demands, and the evidence that would disconfirm it.

#	Gap (evidence)	Direction	Confirm / disconfirm
P1	Strict target 0 (grid), attack channel 56/83	Attack the twin as the objective; study its surface	Reproducible twin-directed attack, released code / primary studies already doing so
P2	Operation 66/83 ($\kappa = 0.32$); design 1, commissioning 0	Measure, then extend, lifecycle coverage	Pre-deployment trust boundaries tested / re-measurement shows coding artefact
P3	Explicit threat model 19/83, adaptive 10/83 vs detection-ML 55/83	Adaptive threat models for learned twin components	Attacker knowledge/capability/objective stated and tested / primary studies already doing so
P4	Poisoning 19, evasion 9 (subst. 7, 1) vs detection 55	Defend learned components	Robust detectors under reactive attack / robust detectors already demonstrated
P5	Code 3/83 (3/49 non-review), data 16/83	Release models, data, attack generators	Third party reruns unaided / corpus stays release-light